Introduction to Cryptography

Lecture 8(A): Multiple-Letter Substitution Ciphers
Playfair Cipher and Hill Cipher with Numericals

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Why Multiple-Letter Ciphers?

Problem with Simple Substitution

In monoalphabetic substitution, each plaintext letter always maps to the same ciphertext letter.

- Single-letter frequency is preserved.
- Common letters such as e, t, a, o remain detectable.
- Frequency analysis can break the cipher.
- Multiple-letter ciphers encrypt groups of letters together.

Key Idea

Instead of encrypting one letter at a time, encrypt digrams or blocks of letters.

Multiple-Letter Substitution



h e l l o $\rightarrow$ he lx lo

Playfair Cipher

Definition

The Playfair cipher is a multiple-letter substitution cipher that treats plaintext digrams as single units and converts them into ciphertext digrams.

- It encrypts two letters at a time.
- It uses a 5×5 matrix of letters.
- The matrix is generated using a keyword.
- I and J are usually combined into one cell.
- It is stronger than simple monoalphabetic substitution.

Playfair Cipher Matrix

For keyword **MONARCHY**, remove repeated letters and then fill remaining letters alphabetically.

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Note

The letters I and J are treated as a single letter to fit 26 letters into 25 cells.

Steps to Construct Playfair Matrix

1. Choose a keyword.
2. Remove duplicate letters from the keyword.
3. Write the remaining keyword letters row-wise in a 5×5 table.
4. Fill the unused alphabet letters in order.
5. Combine I and J into one cell.

Keyword

$$\texttt{MONARCHY} \rightarrow M, O, N, A, R, C, H, Y$$

Preparing Plaintext for Playfair

1. Remove spaces and punctuation.
2. Replace J with I, if required.
3. Divide plaintext into pairs of letters.
4. If repeated letters occur in a pair, insert a filler letter such as X.
5. If the final pair has one letter, append X.

Example

$$\texttt{balloon} \rightarrow \texttt{ba lx lo on}$$

Playfair Encryption Rules

For each plaintext digram:

1. **Same row:** Replace each letter by the letter to its right.
2. **Same column:** Replace each letter by the letter below it.
3. **Rectangle case:** Replace each letter by the letter in its own row but in the column of the other letter.

Wrap Around

In same row or same column, movement is circular.

Rule 1: Same Row

Using the MONARCHY matrix:

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Example

$$\texttt{ar} \rightarrow \texttt{RM}$$

A and R are in the same row. A moves to R, and R wraps around to M.

Rule 2: Same Column

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Example

$$\texttt{mu} \rightarrow \texttt{CM}$$

M and U are in the same column. M moves down to C, and U wraps around to M.

Rule 3: Rectangle Case

M	O	N	A	R
C	H	Y	B	D
E	F	G	I/J	K
L	P	Q	S	T
U	V	W	X	Z

Example

$$\texttt{hs} \rightarrow \texttt{BP}$$

H is replaced by B, and S is replaced by P.

Solved Playfair Numerical 1

Encrypt the plaintext digram:

$$\texttt{ar}$$

using the MONARCHY matrix.

A and R are in the same row:

$$M\ O\ N\ A\ R$$

$$A \to R, \qquad R \to M$$

Answer

$$\texttt{ar} \to \texttt{RM}$$

Solved Playfair Numerical 2

Encrypt the plaintext digram:

$$\texttt{mu}$$

using the MONARCHY matrix.

M and U are in the same column:

$$M, C, E, L, U$$

$$M \to C, \qquad U \to M$$

Answer

$$\texttt{mu} \to \texttt{CM}$$

Solved Playfair Numerical 3

Encrypt the plaintext digram:

$$\texttt{hs}$$

H is at row 2, column 2. S is at row 4, column 4.

$$H \to B \quad \text{and} \quad S \to P$$

Answer

$$\texttt{hs} \to \texttt{BP}$$

Reason

This is the rectangle rule: each letter takes the column of the other letter.

Solved Playfair Numerical 4: Full Word

Encrypt:

`balloon`

using Playfair matrix with keyword MONARCHY.

Prepare plaintext:

`balloon` $\rightarrow$ `ba lx lo on`

- `ba`: rectangle case
- `lx`: rectangle case
- `lo`: rectangle case
- `on`: same row

Solved Playfair Numerical 4: Solution

Using the MONARCHY matrix:

Plain Digram	Cipher Digram
ba	*IB*
lx	*SU*
lo	*PM*
on	*NA*

Ciphertext

`balloon` $\rightarrow$ `IBSUPMNA`

Playfair Decryption Rules

Decryption uses reverse movement:

1. **Same row:** Replace each letter by the letter to its left.
2. **Same column:** Replace each letter by the letter above it.
3. **Rectangle case:** Use the same rectangle rule as encryption.

Example

$$\texttt{RM} \rightarrow \texttt{ar}$$

Security of Playfair Cipher

- It hides single-letter frequency better than monoalphabetic substitution.
- There are 676 possible digrams.
- Digrams are harder to analyze than single letters.
- However, language structure still remains.
- A few hundred letters of ciphertext may be sufficient for cryptanalysis.

Observation

Playfair is stronger than Caesar and monoalphabetic substitution, but not secure by modern standards.

Introduction to Hill Cipher

Definition

The Hill cipher is a multiple-letter substitution cipher that uses matrix multiplication modulo 26.

- Developed by Lester Hill in 1929.
- Plaintext is divided into blocks.
- Each block is treated as a vector.
- Encryption is performed using a key matrix.
- Operations are performed modulo 26.

Letter-to-Number Mapping

a	b	c	d	e	f	g	h	i	j	k	l	m
0	1	2	3	4	5	6	7	8	9	10	11	12

n	o	p	q	r	s	t	u	v	w	x	y	z
13	14	15	16	17	18	19	20	21	22	23	24	25

Hill Cipher Encryption Formula

For a plaintext row vector P and key matrix K:

$$C = PK \bmod 26$$

where:

- P is the plaintext vector.
- K is the key matrix.
- C is the ciphertext vector.
- All calculations are performed modulo 26.

Condition

The key matrix must be invertible modulo 26 for decryption to be possible.

Hill Cipher Decryption Formula

For decryption:

$$P = CK^{-1} \bmod 26$$

where K^{-1} is the inverse of the key matrix modulo 26.

Requirement

$$KK^{-1} \equiv I \pmod{26}$$

- I is the identity matrix.
- If K^{-1} does not exist, the key matrix is invalid.

2×2 Matrix Determinant

For a matrix:

$$K = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The determinant is:

$$\det(K) = ad - bc$$

Invertibility Condition

K is invertible modulo 26 if:

$$\gcd(\det(K), 26) = 1$$

Solved Hill Numerical 1

Encrypt plaintext:

`hi`

using:

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Convert plaintext:

$$h = 7, \quad i = 8$$

$$P = \begin{bmatrix} 7 & 8 \end{bmatrix}$$

Solved Hill Numerical 1: Solution

$$C = PK \bmod 26$$

$$C = \begin{bmatrix} 7 & 8 \end{bmatrix} \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \bmod 26$$

$$C = \begin{bmatrix} 7 \cdot 3 + 8 \cdot 2 & 7 \cdot 3 + 8 \cdot 5 \end{bmatrix} \bmod 26$$

$$C = \begin{bmatrix} 37 & 61 \end{bmatrix} \bmod 26 = \begin{bmatrix} 11 & 9 \end{bmatrix}$$

$$11 = L, \quad 9 = J$$

Ciphertext

$$\texttt{hi} \rightarrow \texttt{LJ}$$

Solved Hill Numerical 2

Encrypt plaintext:

go

using:

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

$$g = 6, \quad o = 14$$

$$P = \begin{bmatrix} 6 & 14 \end{bmatrix}$$

$$C = \begin{bmatrix} 6 & 14 \end{bmatrix} \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \bmod 26$$

Solved Hill Numerical 2: Solution

$$C = \begin{bmatrix} 6 \cdot 3 + 14 \cdot 2 & 6 \cdot 3 + 14 \cdot 5 \end{bmatrix} \bmod 26$$

$$C = \begin{bmatrix} 46 & 88 \end{bmatrix} \bmod 26$$

$$46 \bmod 26 = 20, \qquad 88 \bmod 26 = 10$$

$$20 = U, \qquad 10 = K$$

Ciphertext

$$\text{go} \rightarrow \texttt{UK}$$

Checking Key Matrix Validity

For:

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

$$\det(K) = 3 \cdot 5 - 3 \cdot 2 = 15 - 6 = 9$$

$$\gcd(9, 26) = 1$$

Conclusion

The matrix is invertible modulo 26, so it can be used as a Hill cipher key.

Inverse of 2×2 Matrix Modulo 26

For:

$$K = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

$$K^{-1} = (ad - bc)^{-1} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} \bmod 26$$

- Compute determinant.
- Find multiplicative inverse of determinant modulo 26.
- Form adjugate matrix.
- Reduce all entries modulo 26.

Solved Hill Decryption Numerical

Given:

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}, \qquad C = \begin{bmatrix} 11 & 9 \end{bmatrix}$$

For this key:

$$K^{-1} = \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \bmod 26$$

Decrypt:

$$P = CK^{-1} \bmod 26$$

Solved Hill Decryption: Solution

$$P = \begin{bmatrix} 11 & 9 \end{bmatrix} \begin{bmatrix} 15 & 17 \\ 20 & 9 \end{bmatrix} \bmod 26$$

$$P = \begin{bmatrix} 11 \cdot 15 + 9 \cdot 20 & 11 \cdot 17 + 9 \cdot 9 \end{bmatrix} \bmod 26$$

$$P = \begin{bmatrix} 345 & 268 \end{bmatrix} \bmod 26$$

$$345 \bmod 26 = 7, \qquad 268 \bmod 26 = 8$$

$$7 = h, \qquad 8 = i$$

Plaintext

$$\texttt{LJ} \rightarrow \texttt{hi}$$

Security of Hill Cipher

- It hides single-letter frequency.
- Larger matrices hide more frequency information.
- It can hide two-letter frequency information better than Playfair.
- It is strong against simple ciphertext-only analysis.
- It is vulnerable to known-plaintext attack.

Important

If enough plaintext–ciphertext pairs are known, the key matrix can be recovered.

Playfair vs Hill Cipher

Feature	**Playfair Cipher**	**Hill Cipher**
Unit encrypted	Digram	Block/vector
Mathematical base	5×5 table	Matrix algebra
Key	Keyword	Key matrix
Frequency masking	Good	Better with larger matrix
Weakness	Language structure remains	Known-plaintext attack

Class Practice 1

Using the MONARCHY Playfair matrix, encrypt:

`ea`

Task

Identify the positions of E and A, then apply the Playfair rule.

Class Practice 1: Answer

E is row 3, column 1. A is row 1, column 4.

$$E \to I/J, \qquad A \to M$$

Answer

`ea` $\to$ `IM` or `JM`

Class Practice 2

Using Hill cipher key:

$$K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

Encrypt:

`at`

$$a = 0, \quad t = 19$$

$$P = \begin{bmatrix} 0 & 19 \end{bmatrix}$$

Class Practice 2: Answer

$$C = \begin{bmatrix} 0 & 19 \end{bmatrix} \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix} \bmod 26$$

$$C = \begin{bmatrix} 38 & 95 \end{bmatrix} \bmod 26$$

$$38 \bmod 26 = 12, \qquad 95 \bmod 26 = 17$$

$$12 = M, \qquad 17 = R$$

Answer

$$\texttt{at} \rightarrow \texttt{MR}$$

Quick Quiz

1. What is the main difference between monoalphabetic and multiple-letter substitution?
2. Why are I and J combined in the Playfair matrix?
3. State the three Playfair encryption rules.
4. Encrypt `ar` using the MONARCHY Playfair matrix.
5. Write the Hill cipher encryption formula.
6. What condition makes a Hill cipher key matrix valid?
7. Encrypt `hi` using $K = \begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$.
8. Why is Hill cipher vulnerable to known-plaintext attack?

References

- William Stallings, *Cryptography and Network Security: Principles and Practice*.
- Douglas R. Stinson, *Cryptography: Theory and Practice*.
- Menezes, van Oorschot, and Vanstone, *Handbook of Applied Cryptography*.

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- Clarity of concepts
- Quality of examples
- Pace of the lecture
- Suggestions for improvement

Thank you for your valuable response.



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